Work along a displacement

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate work and identify when a force does no work.

SUCCESS CRITERIA - I can...

1. Identify the force component along displacement.
2. Use meters and newtons to obtain joules.
3. Distinguish effort from mechanical work on an object.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “work” and “displacement” to explain your answer. Sentence starters are optional.

work: energy transferred by a force along a displacement

displacement: change from initial position to final position, with direction

parallel: along the same direction as a line or surface

01 Warm-up: bring back an earlier idea

1. Keep impulse fixed, multiply target mass by **3** and lead time by **5**.
2. **Compare the modeled transverse shift** using an equation.

Word help: lead time = time available before the modeled event

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

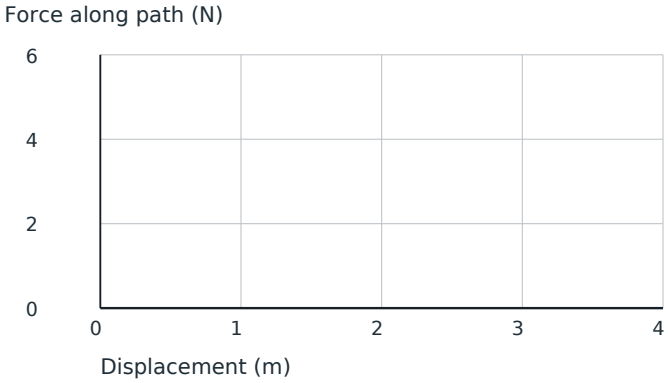
02 Worked example: explain the prediction

Reference: W = Fd.
Constant force parallel to displacement; W is work. No displacement means no work on the object by that force.

Worked reasoning: A constant **5 N** pull acts along a **3 m** displacement. Work is Fd=**15 J**. The object must move for this force to do mechanical work.

Your turn: 1. **Draw the force** and displacement arrows. 2. **Show the 15 J** as area on a force-versus-position graph.

Sentence starter: In my work, I used … to show … . I labeled it … .

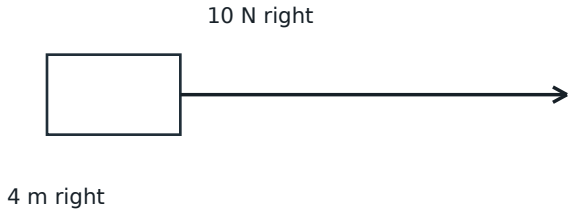


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A box moves **4 m** right while a **10 N** force acts right. **Draw force** and displacement arrows; **predict the sign of work**.

Sentence starter: I predict … . I used … to make this prediction because … .



The rectangle is the box. Add the requested force and displacement labels.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A cart is pulled forward with a constant **18 N** force over **5 m**. A **6 N** friction force opposes the motion.
1. **Draw both forces** and the displacement.
2. **Calculate each force's work** and their net work.

Sentence starter: I found … [unit]. This value means … in this situation.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student holds a heavy bag still. Is the upward support force doing mechanical work on the bag during the hold? Explain without denying the student feels tired.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** object slows from **8** to **4 m/s** in **1.5 s**. **Find impulse** and average net force. Repeat the force calculation for the same velocity change over **3 s**. **Show the signed impulse** on the force–time graph.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found … [unit]. This value means … in this situation.

Force (N)



Time (s)

Marzano check - circle: 1 2 3

07 Exit check

A force does **72 J** over **3 m** parallel to motion.
1. **Find force**.
2. If displacement were zero, what work would that force do?

Sentence starter: I found … [unit]. This value means … in this situation.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

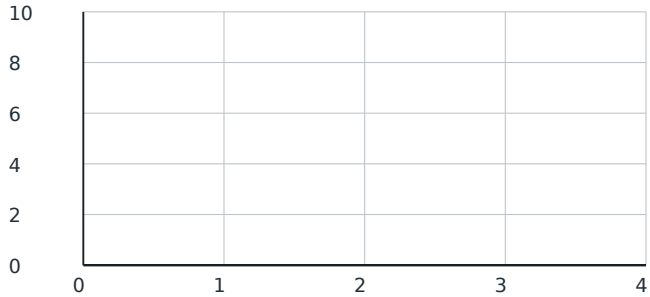
Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A varying force rises linearly from **2** to **8 N** over **4 m**. **Find work** from a graph and compare with using only the final force.

Sentence starter: Both … . They differ in … because … .

Force along path (N)



Marzano check - circle: 1 2 3